Department of Computer Science and Engineering (CSE)
BRAC University

Practice Problem Set 4

CSE251 – Electronic Devices and Circuits

ZENER DIODE

Voltage Regulation, Line or Source Regulation, Load Regulation

Course Description, COs, and Policies

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Midterm and Final Questions

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Problem 1

- A $5\ V$ Zener diode has a knee current of $0.5\ mA$. If it is connected in series with a $10\ V$ supply and a $20\ k\Omega$ resistor, determine whether the Zener will operate on breakdown or not.

Ans: ***Not in breakdown***

Problem 2

- A $9.1\,V$ Zener diode exhibits its nominal voltage at a test current of $28\,mA$. At this current the incremental resistance is specified as $5\,\Omega$. Find the breakdown voltage V_{Z0} of the Zener model. Find the Zener voltage at a current of $10\,mA$ and at $100\,mA$.

Ans: $\boldsymbol{V_{Z0} = 8.96\,\text{V}; 9.01\,V; 9.46\,V}$

Problem 3

- A shunt regulator utilizes a Zener diode whose voltage is $5.1\,V$ at a current of $50\,mA$ and whose incremental resistance is $7\,\Omega$. The diode is fed from a supply of $15\,V$ nominal voltage through a $200\,\Omega$ resistor. What is the output voltage at no load? Find the line regulation and the load regulation.

Ans: $\approx \mathbf{5.1}$ V; $\mathbf{33.8}\ {}^{mV}/_{V}$; $\mathbf{-6.76}\ {}^{mV}/_{mA}$

Problem 4

- A shunt regulator utilizing a Zener diode with an incremental resistance of $5\ \Omega$ is fed through an $82\ \Omega$ resistor. If the raw supply changes by $1.0\ V$, what is the corresponding change in the regulated output voltage?

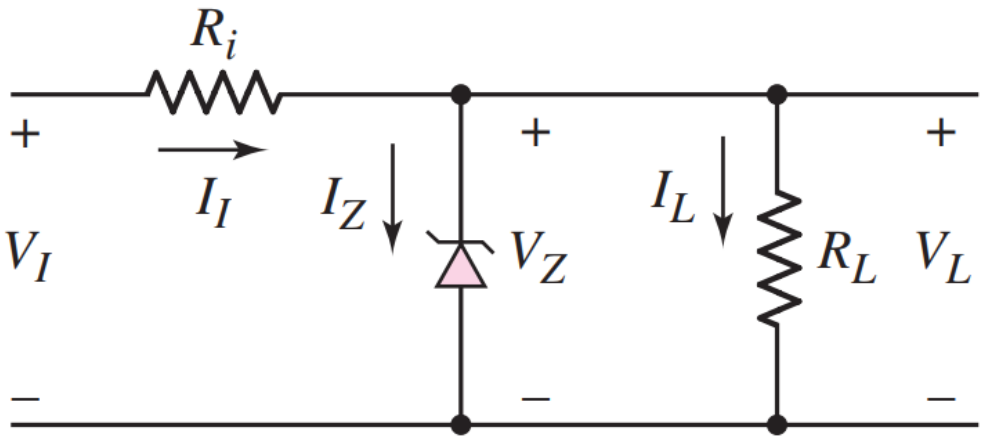
Ans: $\mathbf{58.5\ mV}$

Problem 5

Consider the Zener diode circuit shown below. Let $V_I = 60\,V$, $R_i = 150\,\Omega$, and the breakdown voltage of the Zener $V_{Z0} = 15.4\,V$. Assume the incremental resistance $r_z = 0$. The power rating of the diode is $4\,W$ and the minimum diode current is to be $15\,mA$.

a) Determine the range of diode currents, that is, minimum and maximum allowed diode currents.

b) Determine the range of load resistance so that the diode current is within the range determined in (a).



Ans: (a) $\mathbf{15\,mA \le I_Z \le 259.74\,mA}$
(b) $\mathbf{54.55\,\Omega \le R_L \le 410\,\Omega}$

Problem 6

a) If the Zener diode in the adjacent circuit is to maintain a constant $V_L = 10\ V$ output,

   I. Determine I_L, I_R, and I_Z when $R_L = 180\ \Omega$.

   II. Repeat part I when $R_L = 470\ \Omega$

   III. Determine the value of R_L that will establish maximum power conditions for the Zener diode.

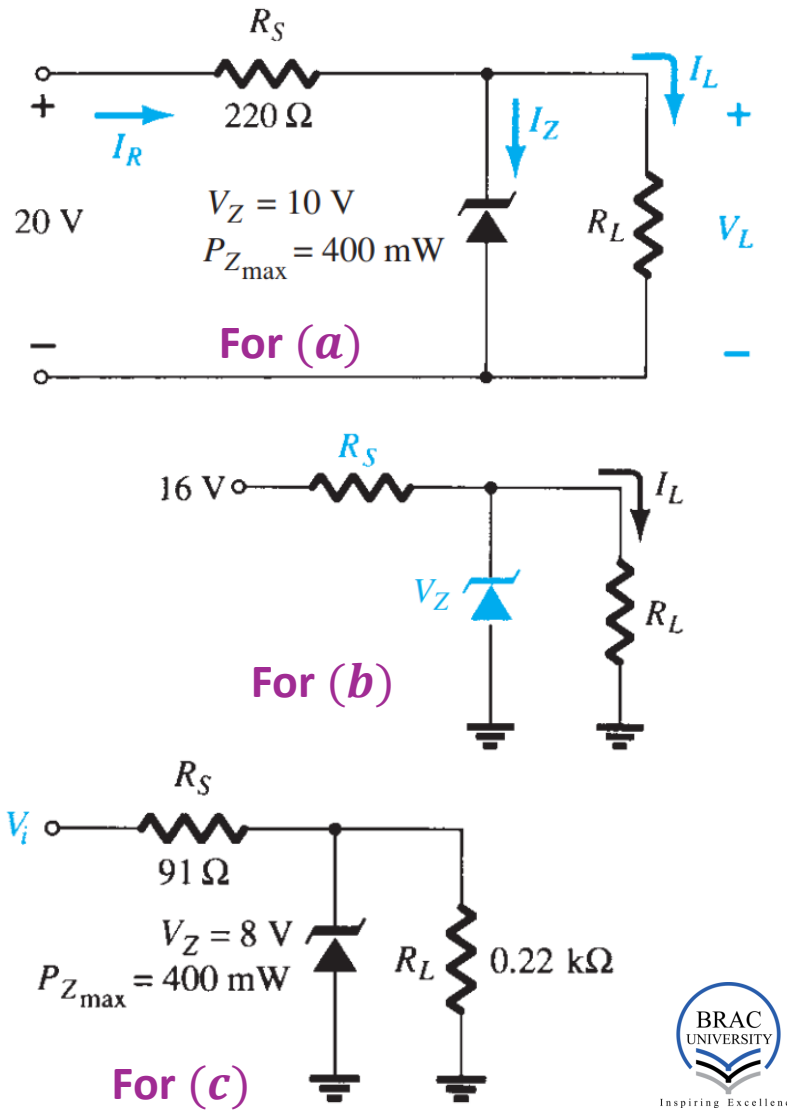
b) If the Zener diode in the adjacent circuit is to maintain a constant V_Z at 12 V for a load variation (I_L) from $0\ mA$ to $200\ mA$. Assume that the knee current $I_{ZK} = 0\ mA$.

   I. Determine a suitable value for R_S.

   II. Determine the maximum power of the diode of part I.

c) If the Zener diode in the adjacent circuit is to maintain a constant $V_Z = 8\ V$ output and not exceed the maximum power rating of the Zener diode, determine the upper limit of V_i.

Ans: (a) $I.\ I_L = I_R = 50\ mA, I_Z = 0\ mA$; $II.\ I_L = 21.28\ mA, I_R = 45.45\ mA, I_Z = 24.17\ mA$; $III.\ 1.83\ k\Omega$; (b) $I.\ R_S = 20\ \Omega$; $II.\ P_{Z_{max}} = 2.4\ W$; $(c)\ 15.86\ V$



Problem 7

The $6.8\,V$ Zener diode in the adjacent circuit is specified to have a voltage of $V_Z = 6.8\,V$ at a current $I_Z = 5\,mA$. The incremental resistance $r_z = 20\,\Omega$ and the knee current $I_{ZK} = 0.2\,mA$. The supply voltage V^+ is nominally $10\,V$ but can vary by $\pm 1\,V$.

a) Find V_O with no load and with V^+ at its nominal value.

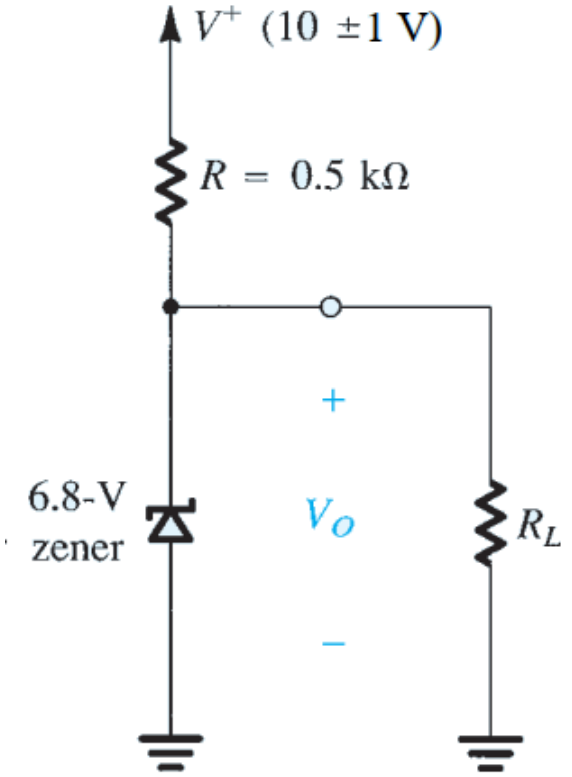
b) Find the line regulation.

c) Find the load regulation.

d) Find the change in V_O when $R_L = 2\,k\Omega$.

e) Find the value of V_O when $R_L = 0.5\,k\Omega$.

f) What is the minimum value of R_L for which the diode still operates in the breakdown region?



Ans: $(a)\ \boldsymbol{V_O = 6.83\ V}$
$(b)\ \frac{\Delta V_O}{\Delta V^+} = \boldsymbol{38.5\ {}^{mV}/_{V}}$
$(c)\ \frac{\Delta V_O}{\Delta I_L} \approx \boldsymbol{-20\ {}^{mV}/_{mA}}$

$(d)\ \boldsymbol{\Delta V_O = -68\ mV}$
$(e)\ \boldsymbol{V_O = 5\ V}$
$(f)\ \boldsymbol{R \approx 1.5\ k\Omega}$

Problem 8

- Design a $7.5\,V$ Zener regulator circuit using a $7.5\,V$ Zener specified at 12 mA. The Zener has an incremental resistance $r_z = 30\,\Omega$ and a knee current of $0.5\,mA$. The regulator operates from a $10\,V$ supply. The value of the input resistor R is $250\,\Omega$. What is the regulator output voltage when the supply is $\pm 10\%$ of its nominal value? What is the output voltage when both the supply is 10% high and the load is removed? What is the smallest possible load resistor that can be used while the Zener operates at a current no lower than the knee current while the supply is 10% low? What is the load voltage in this case?

Ans: $\boldsymbol{V_O = 7.4\,V}$ & $\boldsymbol{7.6\,V}$; $\boldsymbol{V_O = 7.55\,V}$; $\boldsymbol{R_L = 1.04\,k\Omega}$; $\mathbf{V_O = 7.155\,V}$

Problem 9

It is required to design a Zener shunt regulator to provide a regulated voltage of about $10\,V$. The available $10\,V$, $1\,W$ Zener of type 1N4740 is specified to have a $10\,V$ drop at a test current of $25\,mA$. At this current, its incremental resistance r_z is $7\,\Omega$. The raw supply, V_S, available has a nominal value of $20\,V$ but can vary by as much as $\pm 25\%$. The regulator is required to supply a load current of $0\,mA$ to $20\,mA$. Design for a minimum Zener current of $5\,mA$.

a) Find V_{Z0}.

b) Calculate the required value of the series resistance R.

c) Find the line regulation. What is the change in V_O expressed as a percentage, corresponding to the $\pm 25\%$ change in V_S?

d) Find the load regulation. By what percentage does V_O change from the no-load to the full-load condition?

e) What is the maximum current that the Zener in your design is required to conduct? What is the Zener power dissipation under this condition?

Ans: $(a)\ \boldsymbol{V_{Z0} = 9.825\ V}$
$(b)\ \boldsymbol{R = 205.6\ \Omega}$
$(c)\ \boldsymbol{\frac{\Delta V_O}{\Delta V^+} = 32.9\ {}^{mV}/_{V}\ ;\ \%\Delta V_O = \pm 1.646\ or\ 3.29}$

$(d)\ \boldsymbol{\frac{\Delta V_O}{\Delta I_L} \approx -6.77\ {}^{mV}/_{mA}\ ;\ \%\Delta V_O = 1.333}$
$(e)\ \boldsymbol{I_{Z,\,max} = 71.38\ mA;\ P_{Z_{max}} \approx 737\ mW}$

Up Next

Relevant *Questions* from Past Semesters

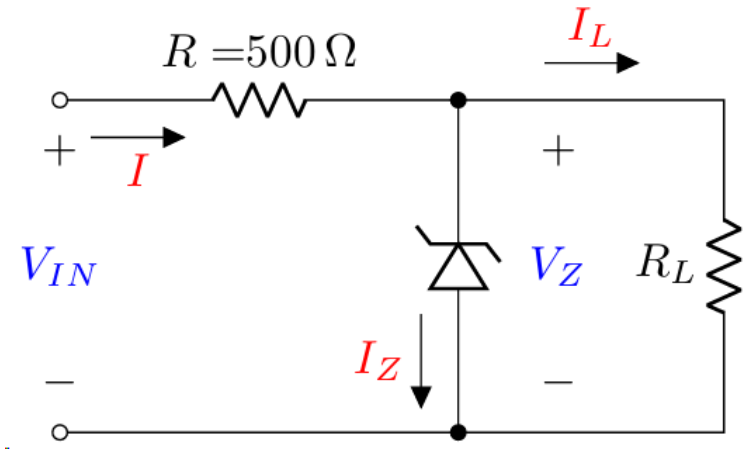
Final – Spring 2024

A Zener diode in the adjacent circuit is specified to have a voltage of $3.8\ V$ when the current through it is $5\ mA$ and a breakdown voltage V_{Z0} of $3.7\ V$. The Zener diode has a Zener knee current of $I_{ZK} = 2\ mA$. The supply voltage V_{IN} is nominally $10\ V$ but can vary by $\pm 1\ V$.

a) Explain in short, why a Zener diode is a better voltage regulator than a regular diode.

b) The breakdown region of a Zener diode is modeled by a constant voltage drop (V_{Z0}) in series with a resistor (r_z). Determine the model parameters r_z for the given Zener diode.

c) Determine the highest load current $I_{L_{max}}$ that the Zener diode can support, above which it fails to function as a regulator. Determine R_L for this case.



Consider that the voltage regulator supply voltage V_{IN} suddenly becomes half of its initial value, i.e., $5\ V$ with a variation of $\pm 0.5\ V$.

d) In this scenario, design the regulator circuit, i.e., find an appropriate value of R so that the Zener diode does not fall out of regulation. Assume that the load found in (c) is connected.

Ans: $(\boldsymbol{b})\ \boldsymbol{r_z = 20\ \Omega}$
$(\boldsymbol{c})\ \boldsymbol{I_{L_{max}} = 12.52\ mA}$; $\boldsymbol{R_L = 298.7\ \Omega}$
$(\boldsymbol{d})\ \boldsymbol{R = 52.3\ \Omega}$

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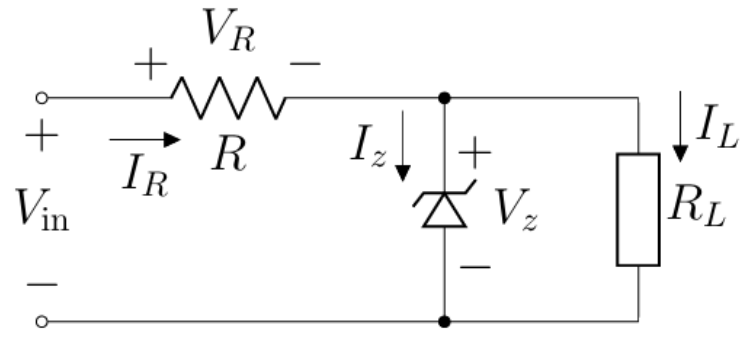
A Zener diode voltage regulator has a regulation voltage of $5.1\ V$ when the current through the Zener diode is $50\ mA$. It is characterized by the parameters, incremental resistance $r_z = 5\ \Omega$, knee current $I_{ZK} = 5\ mA$.

a) Determine the cut-in voltage V_{Z0} for the Zener diode.

b) Assume, the nominal value of the supply voltage, V_{in} is $15\ V$ and input resistance $R = 200\ \Omega$. Calculate the current I_R though R, load resistance R_L, and load current I_L for the worst-case scenario.

c) Now, assume, R_L and R are both $100\ \Omega$ each. Calculate the minimum value of the supply voltage V_{in} at the worst-case scenario.

d) The supply voltage of the regulator is now given as $15 \pm 1\ V$. Design the voltage regulator circuit, i.e., determine the value of R, such that even in the worst-case scenario, voltage regulation is maintained and the circuit sustains a load current of $8\ mA$.

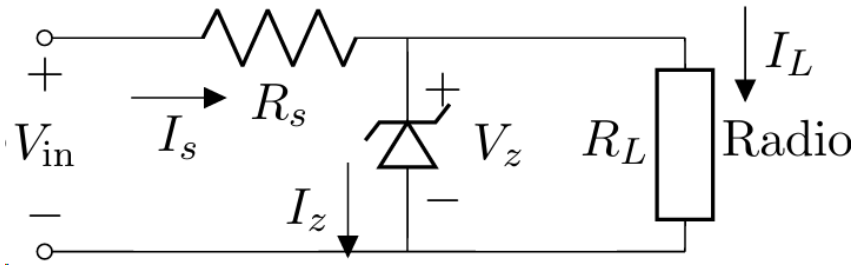


Ans: $(a)\ \boldsymbol{V_{Z0} = 4.85\ V}$
$(b)\ \boldsymbol{I_R = 49.5\ mA;\ R_L = 114.61\ \Omega;\ I_L = 44.5\ mA}$
$(c)\ \boldsymbol{V_{in} = 10.8\ V}$
$(d)\ \boldsymbol{684.62\ \Omega}$

Final – Spring 2023

Adnan was working on a project to build a time machine. He made significant progress on it. While testing, he set the date to March 26, 1971. Surprisingly, the machine worked, and he discovered himself in a house where he met his young grandfather. His grandfather was trying to listen to the radio about the current news in the country. But the radio was not working. Adnan had some knowledge of electronic circuits. So, he opened up the radio and identified that there was a problem with the voltage regulator circuit. Now Adnan needs your help to design the following regulator circuit to help his grandfather.

This circuit functions as a voltage regulator to supply power to the radio. The radio operates at $12\ V$. A battery bank of pencil batteries (represented by V_{in}) supplies the power, which can have a variable voltage range of $15\ V$ to $18\ V$. The current required by the radio (I_L) varies between $0\ mA$ (when turned off) and $10\ mA$ (at full volume). A Zener diode, with a specified breakdown voltage of, $V_{Z_0} = 12\ V$, is used in the circuit. It is characterized by the parameters, incremental resistance $r_z = 0.1\ k\Omega$, knee current $I_{zk} = 2\ mA$.



a) Identify the worst-case conditions and calculate the Zener current (I_z), Zener voltage (V_z), the input voltage (V_{in}), and the load current (I_L) in this worst-case scenario.

b) Calculate the current I_s, the voltage V_s (across R_s) and the resistor R_s in the worst-case scenario.

c) Design the circuit, i.e., find the value of R_L, such that even in the worst-case scenario, voltage regulation is maintained.

Ans: $(a)\ \boldsymbol{I_z = 2\ mA; V_Z = 12.2\ V;\ V_{in} = 15\ V; I_L = 10\ mA}$
$\boldsymbol{(b)\ I_s = 12\ mA; V_s = 2.8\ V; R_s = 233.33\ \Omega}$
$\boldsymbol{(c)\ R_L = 1220\ \Omega}$

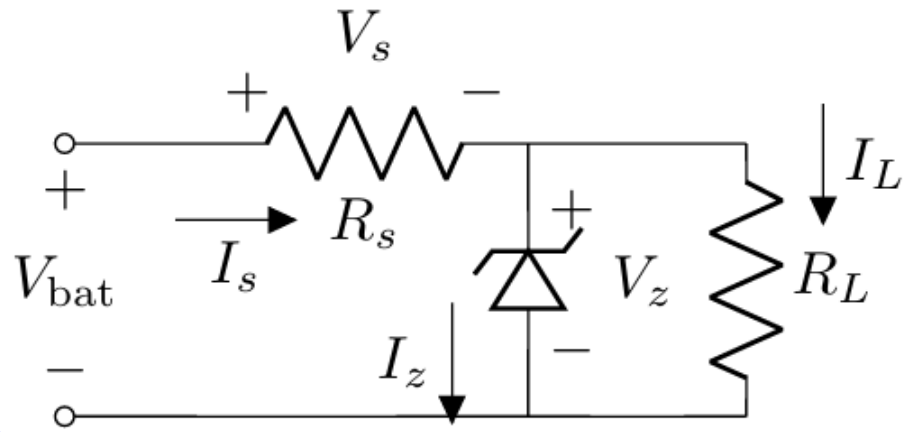
Final – Fall 2022

The circuit below is a voltage regulator used to power a load R_L from a battery V_{bat}. For this circuit, $R_s = 0.2\,k\Omega$, $R_L = 2\,k\Omega$, breakdown voltage of the Zener diode $V_{Z0} = 4\,V$, incremental resistance $r_z = 0\,\Omega$, and knee current $I_{zk} = 0.5\,mA$.

a) Identify the Zener current I_z in the worst-case scenario. In this worst-case scenario, calculate the Zener voltage V_z, load current I_L and input current I_s.

b) Design the circuit, i.e., find the minimum value of the input voltage V_{bat} such that, voltage regulation is maintained even in the worst-case scenario.

c) Determine whether the circuit will maintain regulation if V_{bat} is increased. If yes, argue if it should be increased or not.



Ans: (a) $I_z = 0.5\,mA$; $V_Z = 4\,V$; $I_L = 2\,mA$; $I_s = 2.5\,mA$
(b) $V_{bat} = 4.5\,V$
(c) **Will be maintained**

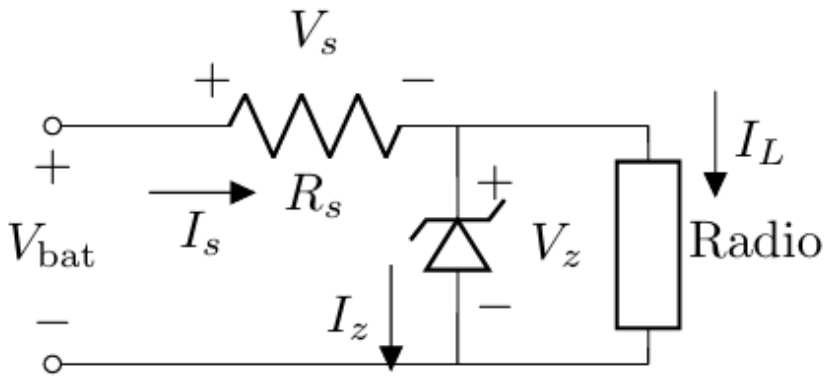
Mid – Summer 2022

The circuit below is a voltage regulator used to power a car radio (which requires $\approx 9\,V$) from the car battery, V_{bat} whose voltage may vary between $11\,V$ and $13.6\,V$. The current in the radio, I_L, will vary between 0 (OFF) to $9\,mA$ (full volume). The Zener diode in the circuit is specified with parameters, breakdown voltage $V_{Z0} = 9\,V$, incremental resistance $r_z = 0.05\,k\Omega$, and knee current $I_{zk} = 1\,mA$.

a) Identify the worst-case conditions and calculate the Zener current (I_z), Zener voltage (V_z), the input voltage (V_{bat}), and the load current (I_L) in this worst-case scenario.

b) Calculate the current (I_s) and the voltage (V_s) of the input resistor R_s in the worst-case scenario.

c) Calculate the line regulation for this circuit.



Ans: (a) $\boldsymbol{I_z = 1\,mA; V_Z = 9.05\,V; V_{bat} = 11\,V; I_L = 9\,mA}$
(b) $\boldsymbol{I_s = 10\,mA; R_s = 0.195\,k\Omega}$
(c) $\boldsymbol{25\ {}^{mV}/_{V}}$

Acknowledgement and References

Some of the problems in this set are taken or adapted from the following sources:

1. Sedra, A. S., & Smith, K. C., Microelectronic Circuits, Oxford University Press
2. Neamen, D. A., Microelectronics: Circuit Analysis and Design, McGraw-Hill

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